Mudit Garg
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Research : ADS Library

Research Interests and Expertise

I am mainly interested in understanding the **co-evolution of inspiraling Black Hole Binaries (BHBs) with their galactic nuclei** via **multi-messenger astronomy** facilitated by the Gravitational-Wave mission LISA, and optical surveys LSST/Roman. For this purpose, I use pen-and-paper, semi-analytical, and numerical techniques. I have shown for the first time, using realistic Bayesian inference and state-of-the-art hydrodynamical simulations, that **gas-induced effects and orbital eccentricity are essential to include when modeling massive BHBs to extract unprecedented astrophysical information.**

Education

08/'21 – 10/'25
Doctor of Philosophy in Gravitational Wave Astrophysics
University of Zurich — Advisor: Prof. Lucio Mayer
Decoding the Sub-Parsec Scale Environment of LISA Massive Black Hole Binaries with Gravitational Waves

09/'18 – 12/'20
Master of Science in Physics **with distinction** — GPA: 5.87/6
ETH Zurich — Thesis supervisor: Prof. Lavinia Heisenberg
Relativistic, ghost-free, and covariant hybrid model for MOND: f(Q)

07/'14 – 06/'18
Bachelor of Technology in Engineering Physics — GPA: 8.15/10
IIT Delhi — Thesis supervisor: Prof. Ajit Kumar
Geodesics near a charged black hole in $\left(R \pm \mu^4/R\right)$ *gravity*

Academic Experience

11/'25 – *10/'27*
Swiss Mobility Postdoctoral Fellow in Astrophysics
New York University — Host: Prof. Andrew MacFadyen

Selected Talks - 9 seminars, 4 invited + 10 contri. conferences, and 8 individual

Seminars at Columbia University — NYC
10/'26 *THEA: TBD*
09/'24 Astro: *Decoding Astrophysics from inspiraling LISA MBHBs*

CfA High Energy Seminar at Harvard University — Boston
09/'26 *TBD*

GRAPPA talks at University of Amsterdam — Amsterdam
09/'26 *Colloquium: TBD*
06/'24 *Seminar: Astrophysical signatures on the LISA data stream from MBHBs*

10/'24 **CTC Theory Seminar** at University of Maryland [25+25 minutes] — College Park
Decoding Astrophysics from inspiraling LISA MBHBs

09/'24 **Astrophysics Seminar** at Johns Hopkins University [45+15 minutes] — Baltimore
Decoding Astrophysics from inspiraling LISA MBHBs

05/'24 **Cosmology Seminar** at Max Planck Institute for Astrophysics — Garching
Astrophysical signatures on GWs from LISA MBHBs

02/'24 **DAMTP GR Seminar** at University of Cambridge [50+10 minutes] — Cambridge
Astrophysical signatures on the LISA data stream from MBHBs

Monday Afternoon Talks at MIT Kavli Institute — Boston
09/'26 *TBD*
09/'24 *Decoding Astrophysics from inspiraling LISA MBHBs*

	LISA Symposium
06/'26	*Chaotic migration of LISA EMRIs in a turbulent accretion disk* — College Park
07/'24	*Poster: Astrophysical signatures on the LISA data stream from MBHBs* — Dublin
05/'26	**25$^{\mathrm{th}}$ Eastern Gravity Meeting** at Cornell University — Ithaca *Chaotic migration of LISA EMRIs in a turbulent gas flow*
06/'25	**Workshop**: Astrophysical Dynamics: from planets, to stars, to black holes — Copenhagen Niels Bohr Institute — *Characterizing sub-pc environment of LISA MBHBs*
06/'25	**Conference (Invited)**: DYNAMIX — Cambridge Institute of Astronomy, Cambridge — *Characterizing sub-pc environment of LISA MBHBs*
05/'25	**Workshop (Invited)**: Gravitational Wave Probes of Black Hole Environments — Trieste IFPU, SISSA & ICTP — *Characterizing sub-pc environment of LISA MBHBs*
03/'25	**Workshop (Invited)**: Frontiers of Astrophysical Black Holes — Sexten Sexten Center for Astrophysics — *What solves the 'final parsec' problem for LISA MBHBs?*
	LISA Astrophysics Working Group Meeting
11/'24	*What solves the 'final parsec' problem for LISA MBHBs?* — Garching
09/'23	*The minimum measurable eccentricity from GWs of LISA MBHBs* — Milan
08/'24	**Conference (Invited)**: New ideas on the origin of BH mergers — Copenhagen Niels Bohr Institute — *Astrophysical signatures on the LISA data stream from MBHBs*
07/'23	**Conference:** GW populations: what's next? at University of Milano-Bicocca — Milan *The measurability of gas and eccentricity from GWs of LISA MBHBs*
11/'22	**Conference:** LISA data analysis: classical methods to machine learning — Toulouse CNRS, L2IT, APC, CEA, and CNES — *The imprint of Gas on GWs from LISA IMBHBs*
09/'22	**Conference:** Origin, growth and feedback of BHs in dwarf galaxies — San Sebastian Donostia International Physics Center — *The imprint of Gas on GWs from LISA IMBHBs*
05/'22	**Conference:** IMBHs: New Science from Stellar Evolution to Cosmology — San Juan CIERA, Northwestern University — *Gas impact on GWs from LISA IMBHBs*
10/'24	**CIERA theory group meeting** at Northwestern University — Evanston *Decoding Astrophysics from inspiraling LISA MBHBs*
09/'24	**Branch Lunch** at NASA Goddard — Greenbelt *Decoding Astrophysics from inspiraling LISA MBHBs*
09/'24	**Astro Coffee** at Institute of Advanced Study — Princeton *Measuring eccentricity and gas-induced perturbation from GWs of LISA MBHBs*

Research visits

11/'25 –	**Center for Computational Astrophysics (CCA), Flatiron Institute** — Frequent Visitor
06/'25	**Niels Bohr International Academy, University of Copenhagen** *Host: Prof. Johan Samsing* — Copenhagen
11/'23	**Max Planck Institute for Gravitational Physics (Albert Einstein Institute)** *Host: Dr. Jonathan Gair* — Potsdam

Programs/Schools

09/'26 | **Gravitational-Wave Eccentricity workshop** *Corfu Summer Institute* Corfu

07/'25 | **MIAPbP Program**: Enabling Future GW Astrophysics in the Milli-Hertz Regime
Munich Institute for Astro-, Particle and BioPhysics Garching

09/'24 | **Workshop**: Fundamental Physics Meets Waveforms With LISA
Max Planck Institute for Gravitational Physics (Albert Einstein Institute) Potsdam

09/'23 | **Kavli-Villum School:** Gravitational Waves *Corfu Summer Institute* Corfu

11/'22 | **Workshop:** LISA data analysis: classical methods to machine learning
CNRS, L2IT, APC, CEA, and CNES Toulouse

01/'22 | **Saas-Fee School:** Multi-Messenger GW Astronomy
Swiss Society for Astrophysics and Astronomy Saas-Fee

08/'21 | **NBIA School:** Gravitational wave astrophysics
Niels Bohr Institute, University of Copenhagen Copenhagen

Professional responsibilities and memberships

'26 – | Co-organizer of the Astrophysics Seminars at *Department of Physics, New York University*

'25 – | Referee for ApJ, PRD

'26 – | Contributor to the LISA Waveform parameter space for astrophysical sources and updates to the MBHB astrophysics white paper core projects

'21 – | Core member of the LISA consortium and its astrophysics, waveforms, and data challenge working groups, and its early career scientist group (LECS)

'23 – '25 | Organizer of the 'GWs, BHs, and Compact Binaries' Seminars *University of Zurich*

'21 – '25 | Teaching assistant for several astrophysics courses at the University of Zurich

'21 | Research assistant at the Department of Astrophysics, University of Zurich

Skills

Software and programming language

- GIZMO: Performed and analyzed hydro simulations of gravitational-wave driven LISA massive black hole binaries embedded in accretion disk
- LISABETA: Added several waveform modules. Also, experienced user and frequently edits it to suit a given project's needs. It provides a complete LISA response and Bayesian inference primarily using the PTMCMC sampler.
- ATHENA++: Performing (magneto-) hydrodynamical simulations of gas-embedded LISA binaries
- ERYN: I have performed reversible jump MCMC with this sampler in LISABETA.
- MATHEMATICA: Frequent user to do analysis and plotting.
- PYTHON: I mainly use this programming language to perform analysis and make plots.

Languages: English | Hindi
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